

Ethanol Blending in Gasoline – Spain

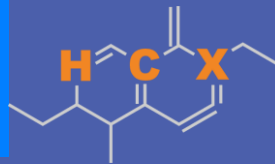
September 2025



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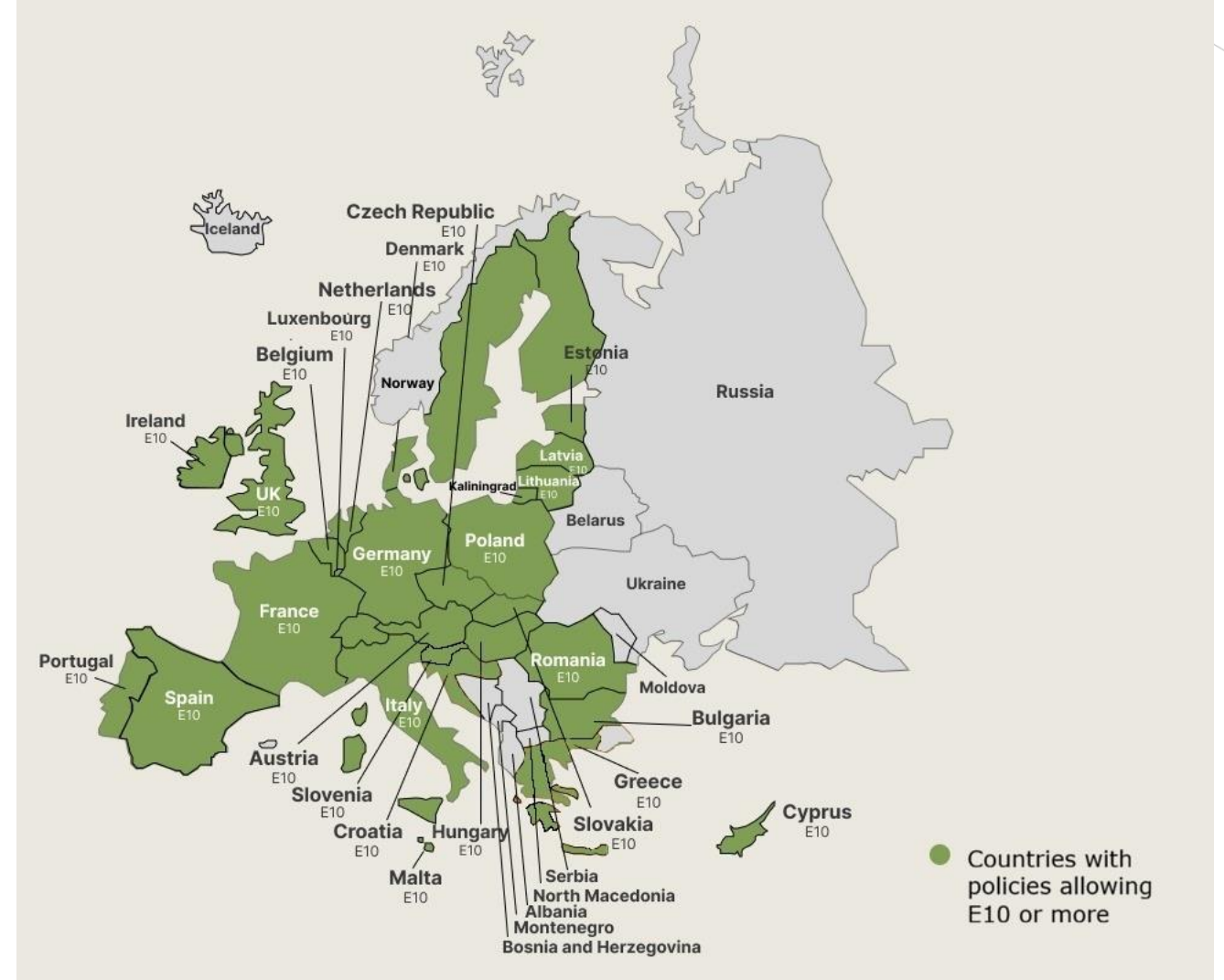


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Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Spain

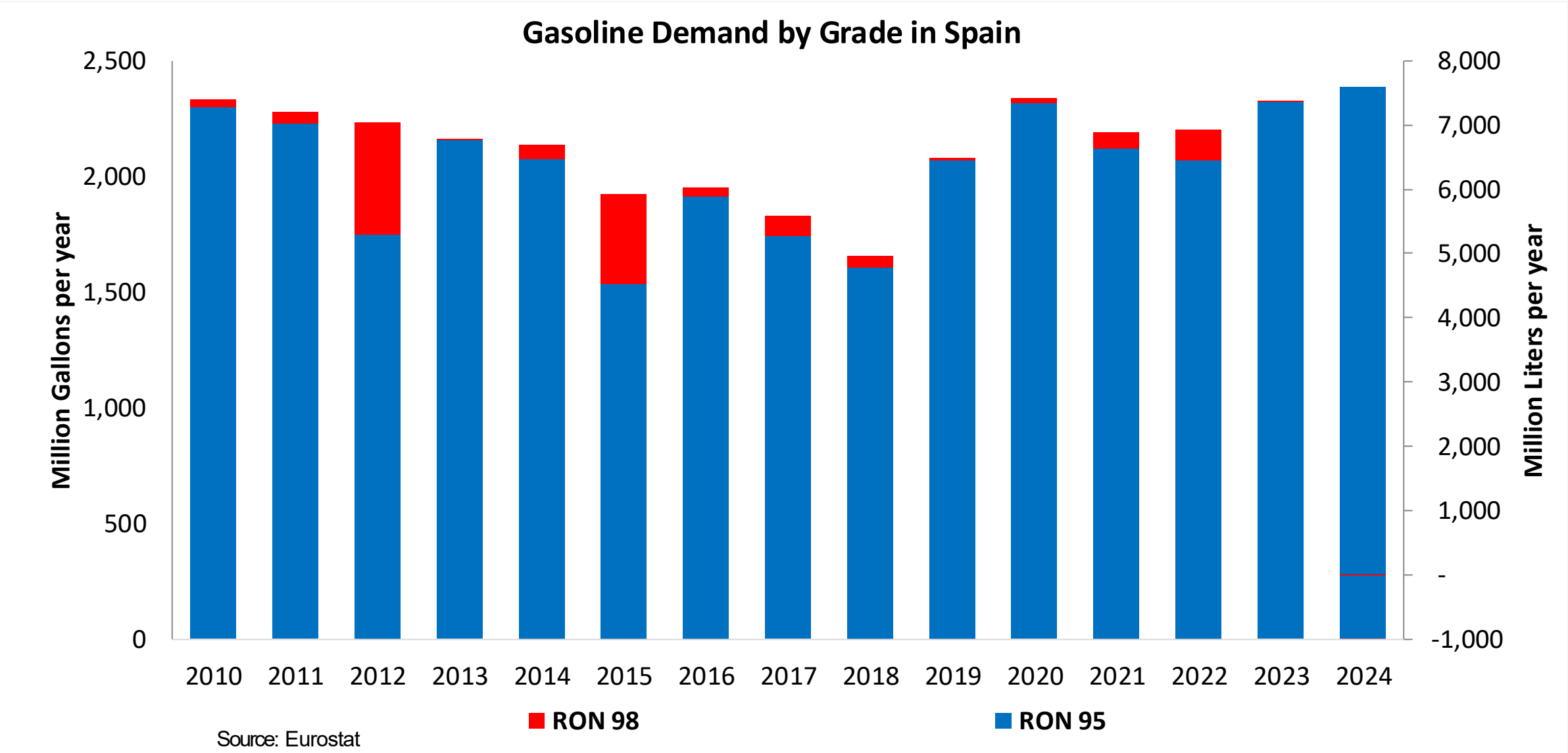


In 2024, gasoline consumption in Spain was 7,577 million liters (2,002 million gallons) and growth rate was 3.1%. Gasoline production and net imports were 11,890 and -3,071 million liters (3,141 and -811 million gallons) correspondingly. Spain complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.0 RON.

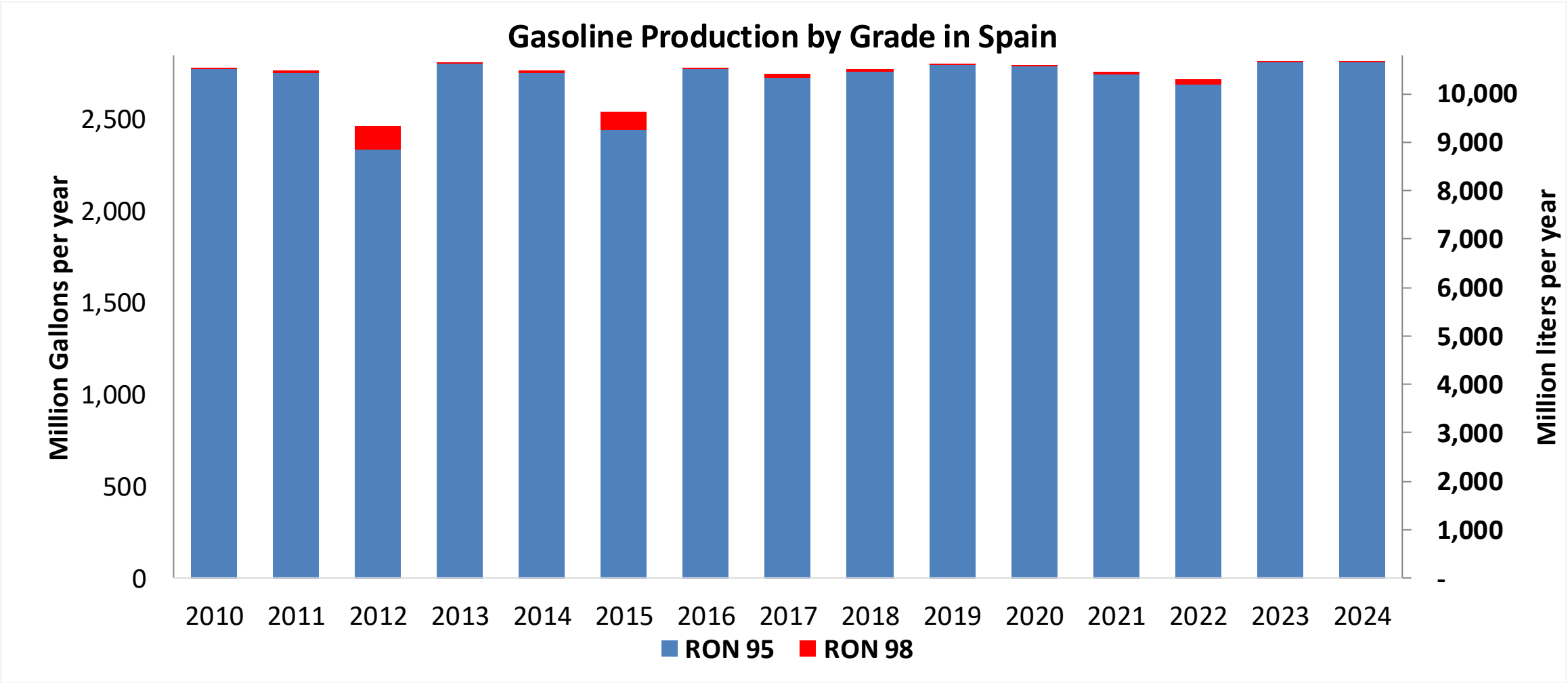
In 2024, ethanol consumption in Spain was 253 million liters (67 million gallons) representing 3.3% of gasoline demand. Ethanol production and Net Imports were 545 and -292 million liters (67 and -77 million gallons) correspondingly. Ethanol sources are Corn 60%, Cereals 25% and Sugar Beet/Wine 40%.

Source: Eurostat, European Commission

Gasoline Demand in Spain

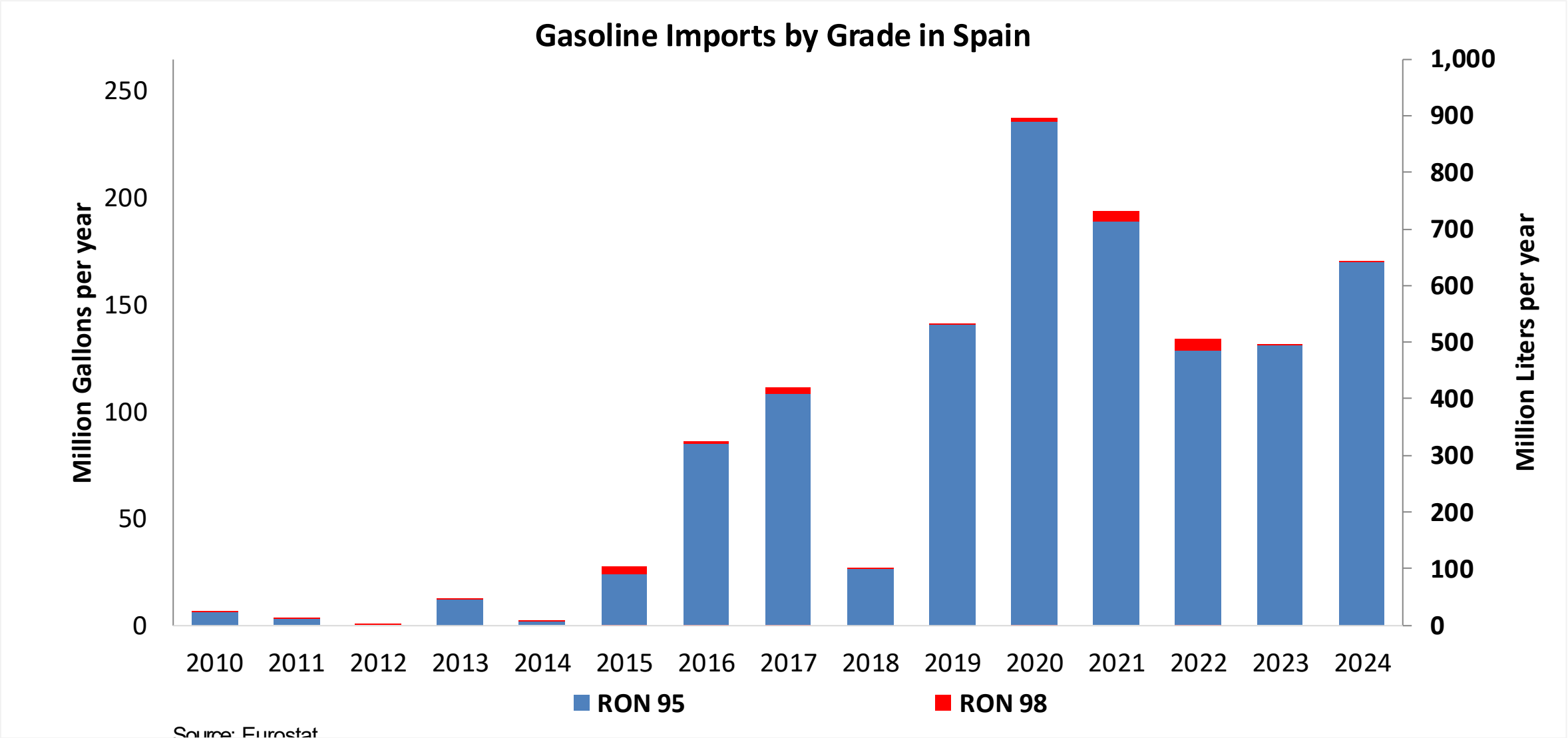


Gasoline Production in Spain

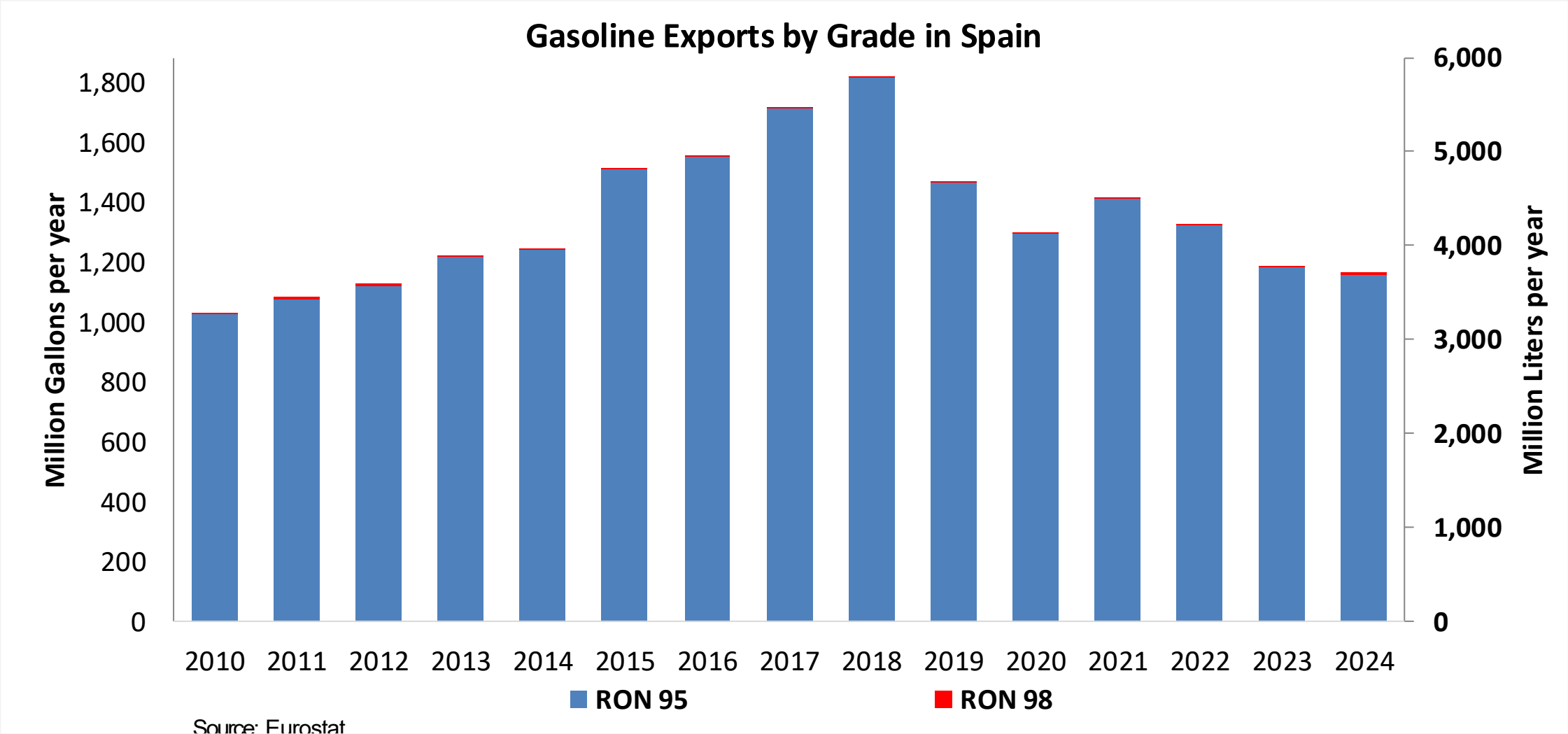


Source: Eurostat

Gasoline Imports in Spain

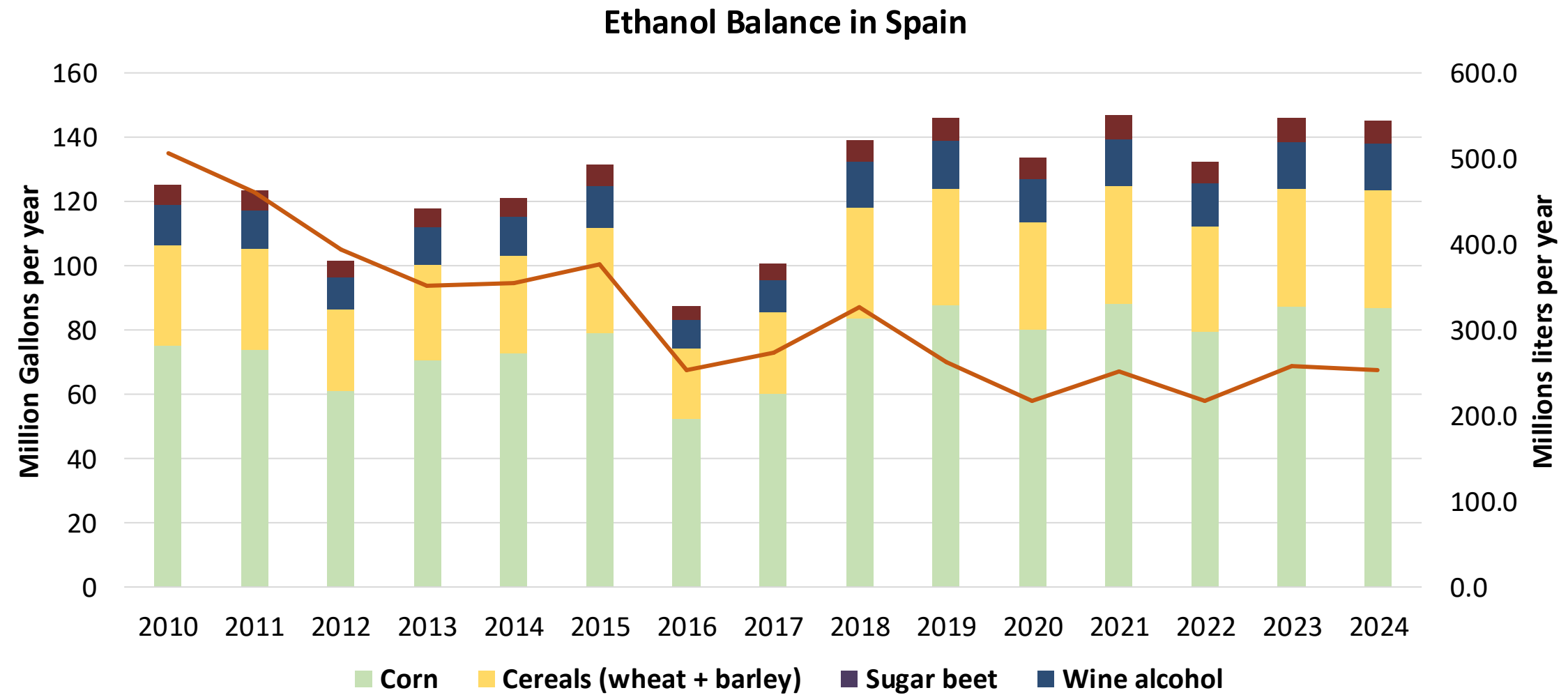


Gasoline Exports in Spain



Source: Eurostat

Ethanol Balance in Spain



Source: Eurostat

Gasoline Quality in Spain

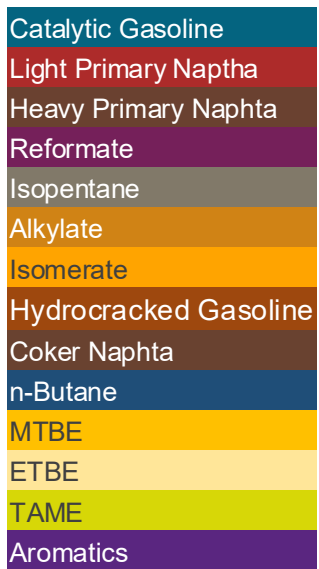
Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	11.5							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexene ring. The structure features a double bond between carbons 1 and 2, with substituents at various positions: a hydrogen atom (H) at C1, a methyl group (CH3) at C2, a methyl group (CH3) at C3, a methyl group (CH3) at C4, a methyl group (CH3) at C5, and a methyl group (CH3) at C6. The carbons are labeled with orange letters: H, C, X, and C.

0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%



Source: Faro90

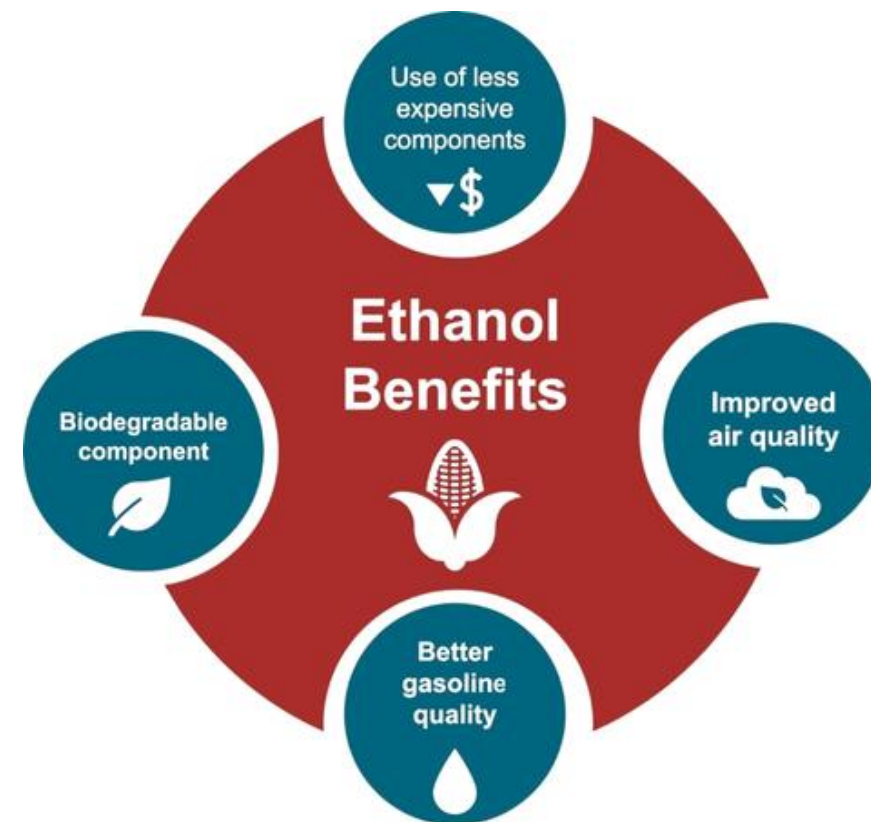
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

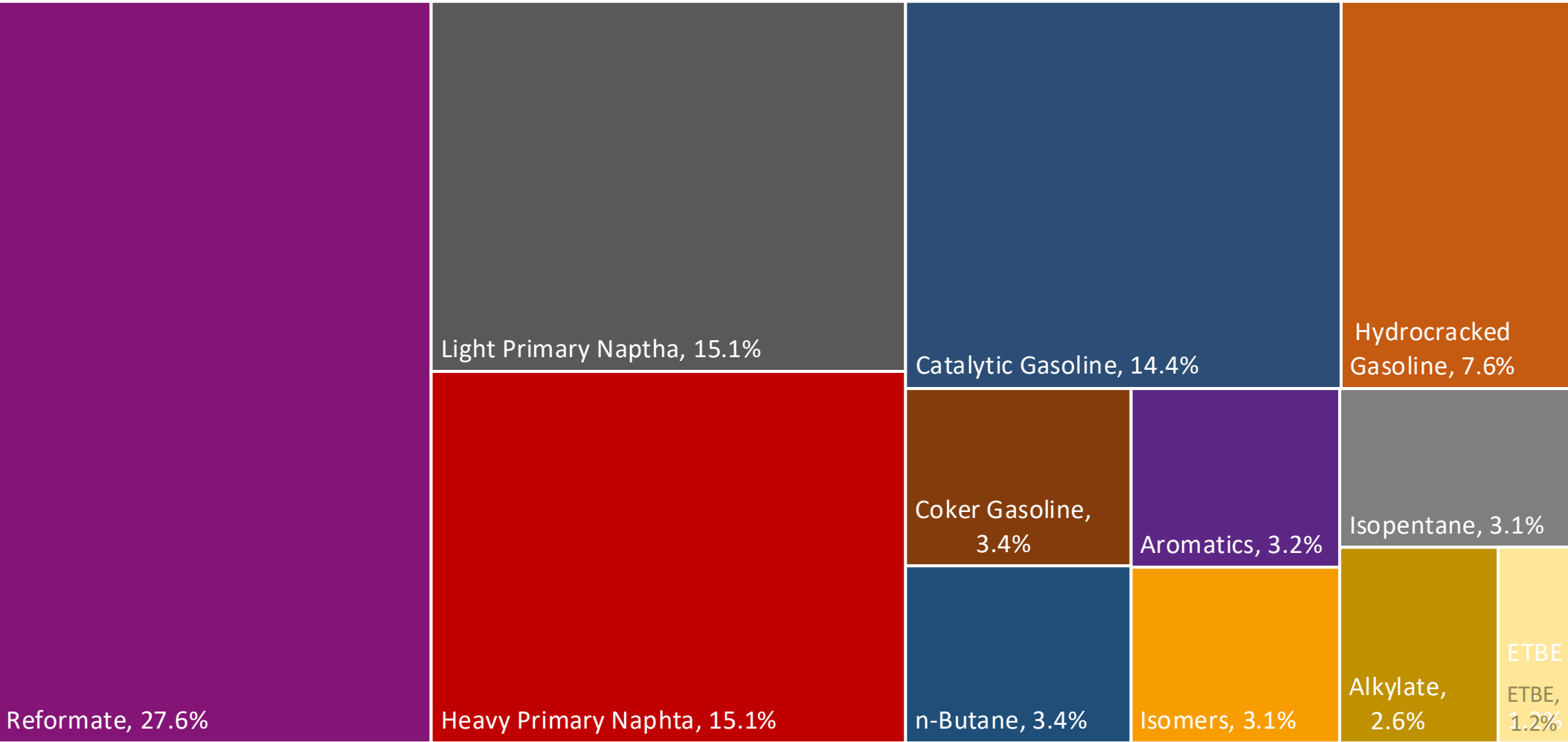
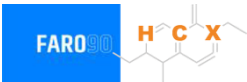
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

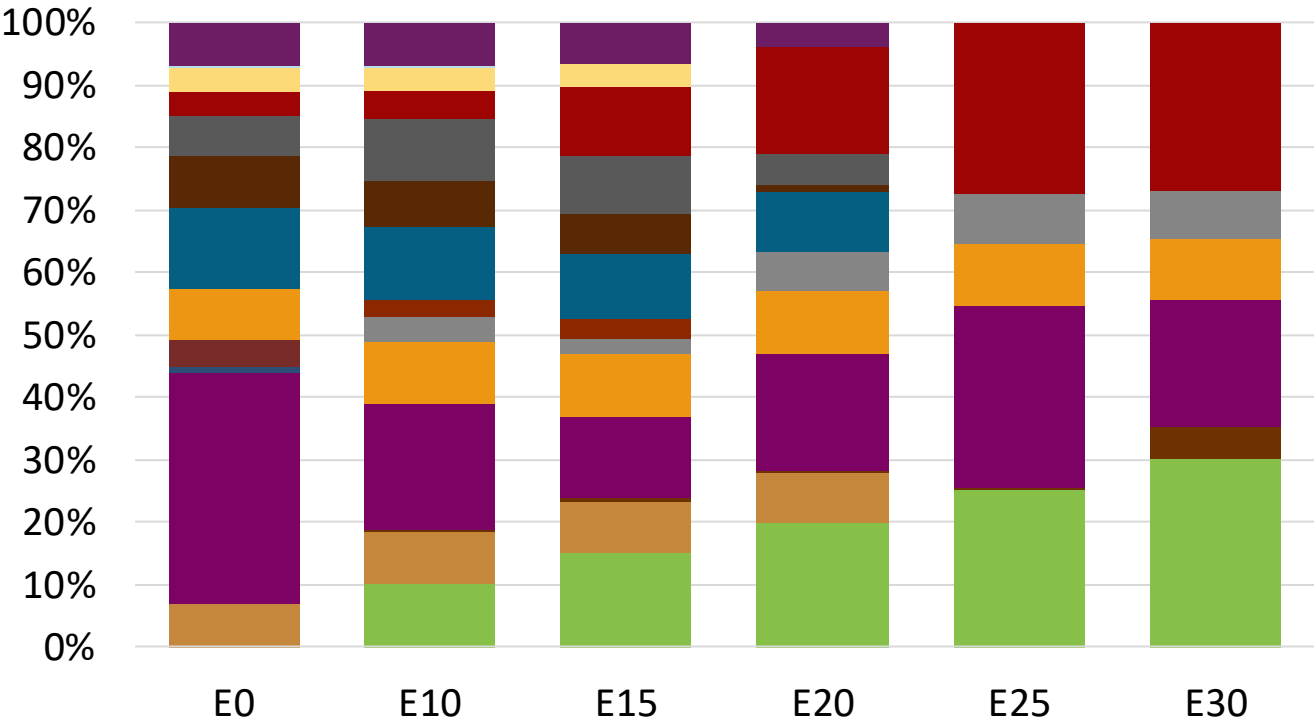
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Spain Available Blending Components



Spain – Gasoline 95 – Constant Octane

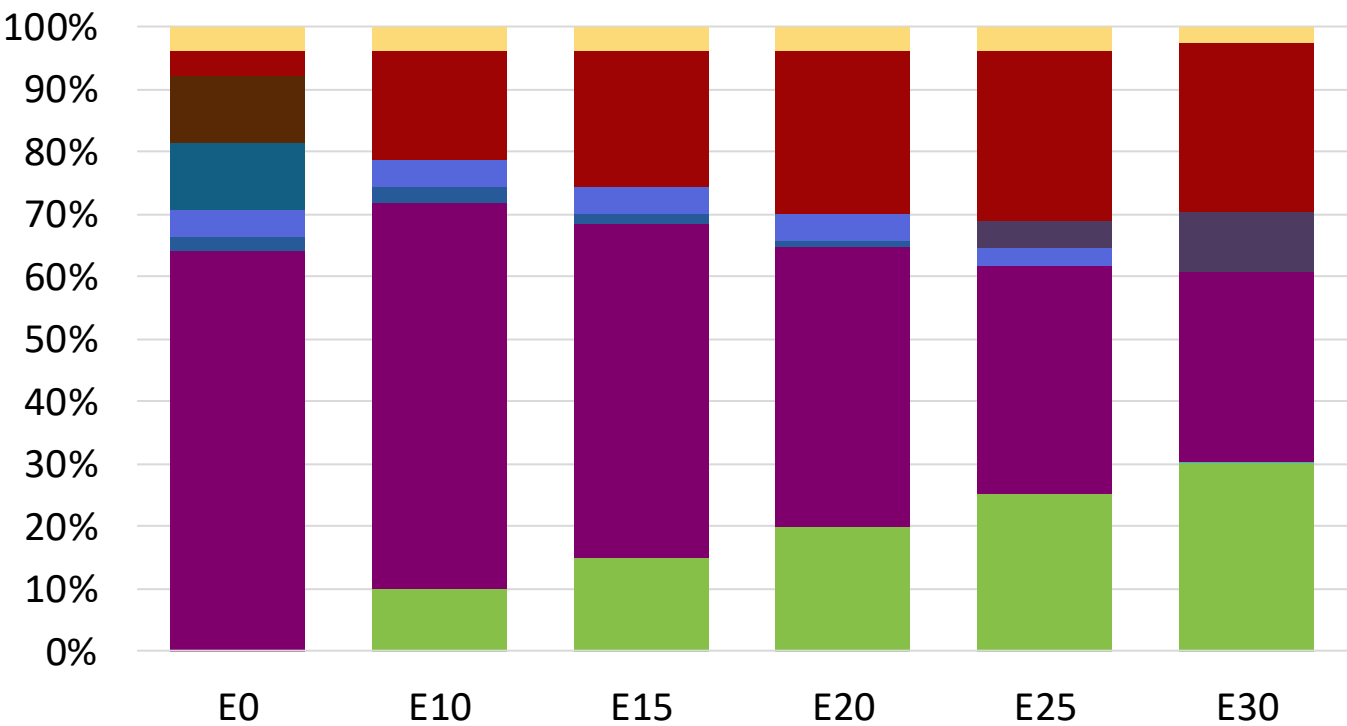


Average CIF 2024
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.3	95.8
Price (US\$/gal)	\$ 2.280	\$ 2.168	\$ 2.088	\$ 1.998	\$ 1.883	\$ 1.820

Source: Faro90

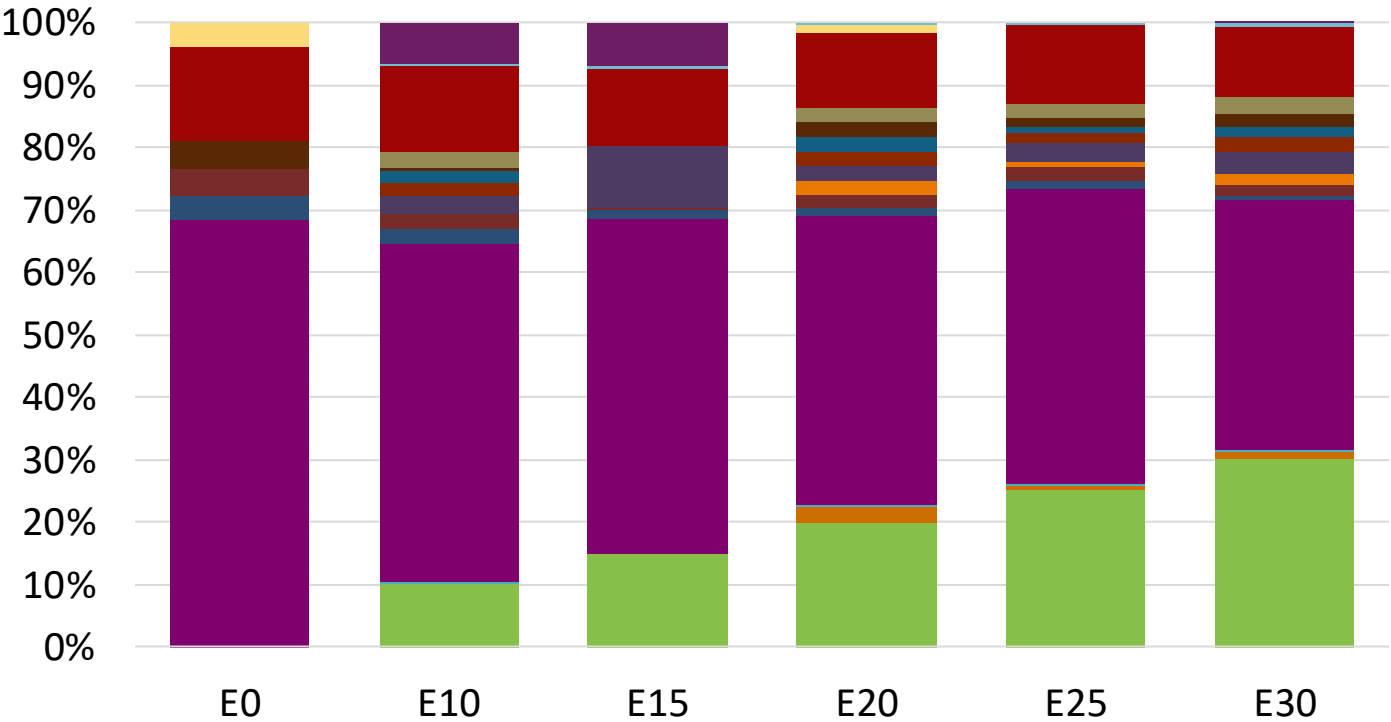
Spain - Gasoline 98 - Constant Octane



Average CIF 2024
Prices

Octane (RON)	98.0					
Price (US\$/gal)	\$ 2.349	\$ 2.225	\$ 2.156	\$ 2.081	\$ 2.003	\$ 1.924

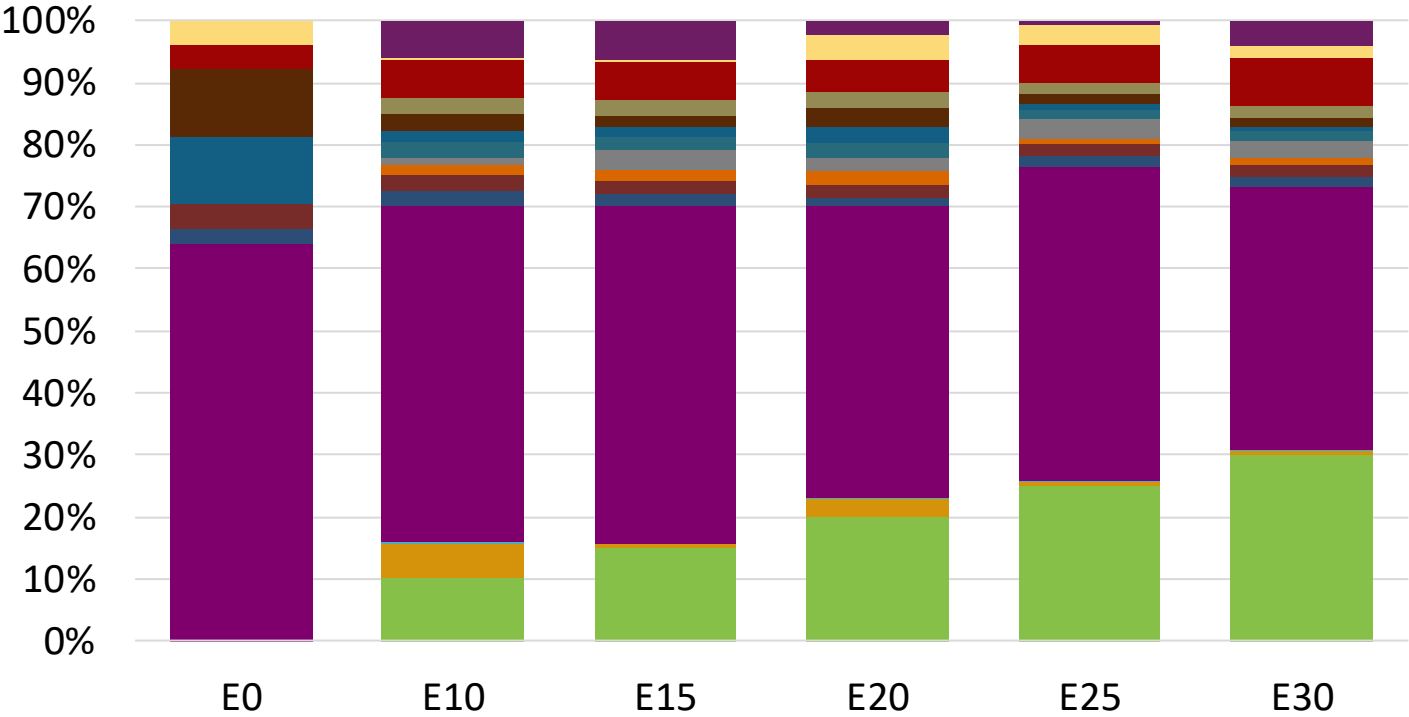
Spain – Gasoline 95 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	95.0	96.8	98.2	99.9	101.4	102.6
Price (US\$/gal)	\$ 2.212	\$ 2.212	\$ 2.212	\$ 2.212	\$ 2.212	\$ 2.212

Spain – Gasoline 98 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	98.0	99.3	100.9	102.5	104.0	105.0
Price (US\$/gal)	\$ 2.349	\$ 2.349	\$ 2.349	\$ 2.349	\$ 2.349	\$ 2.349

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

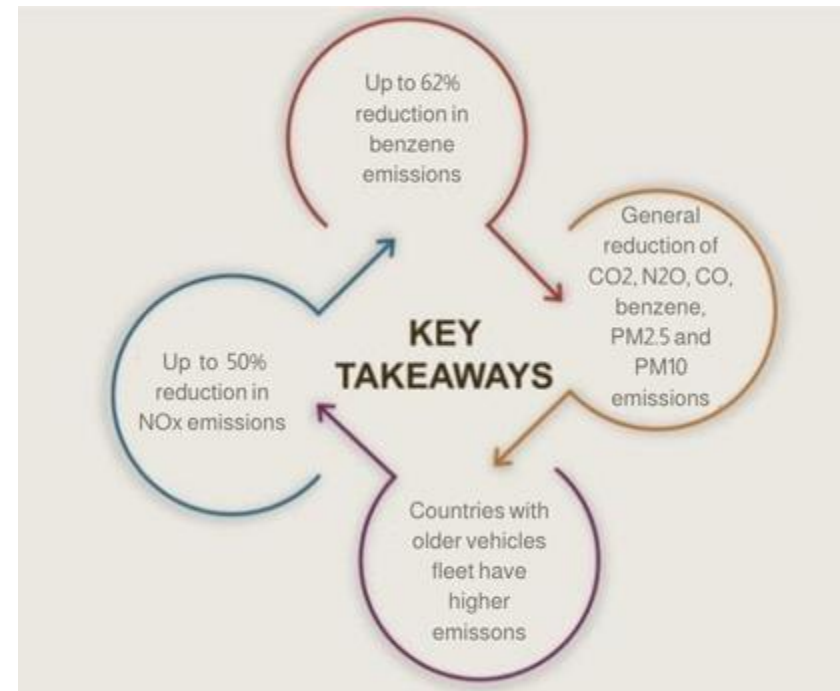
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

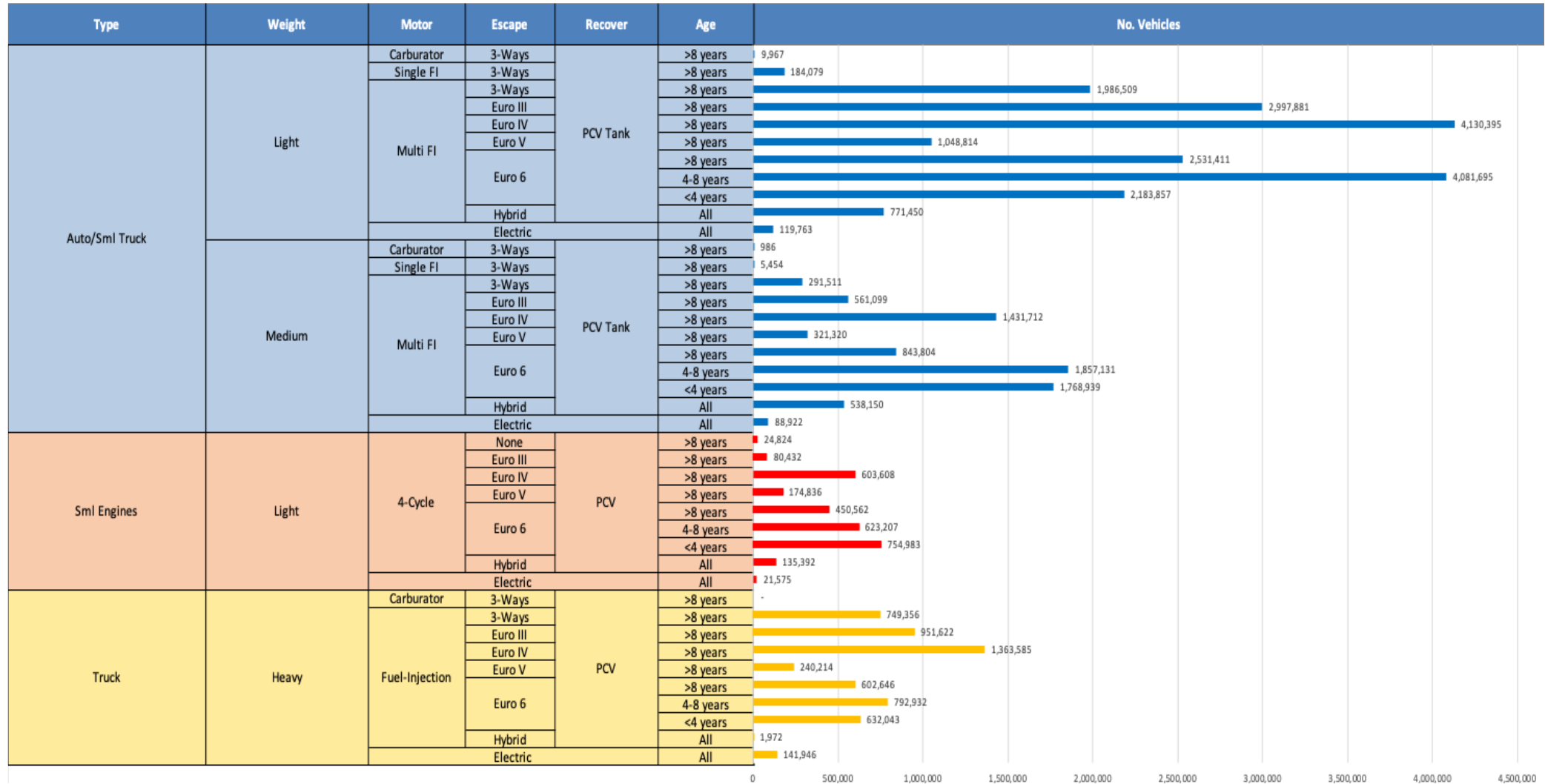
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - Spain

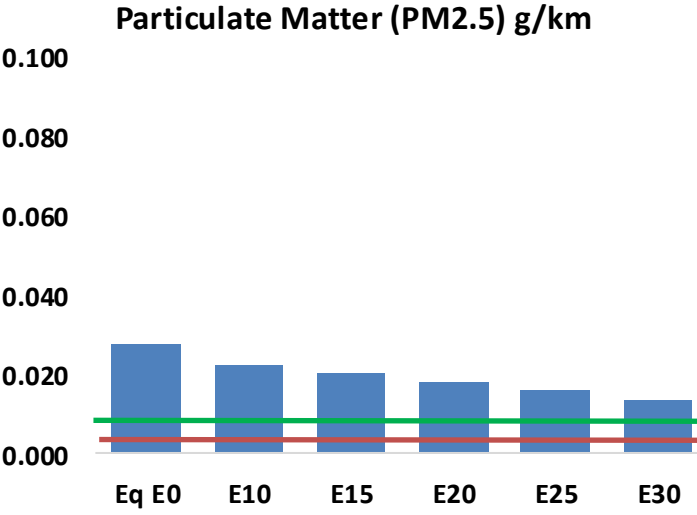
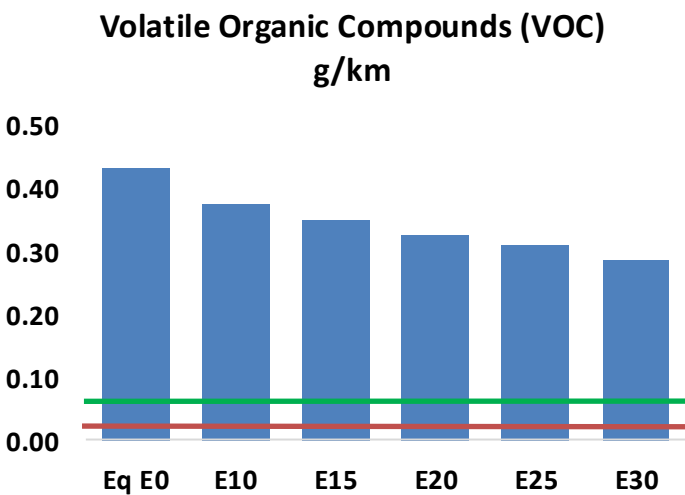
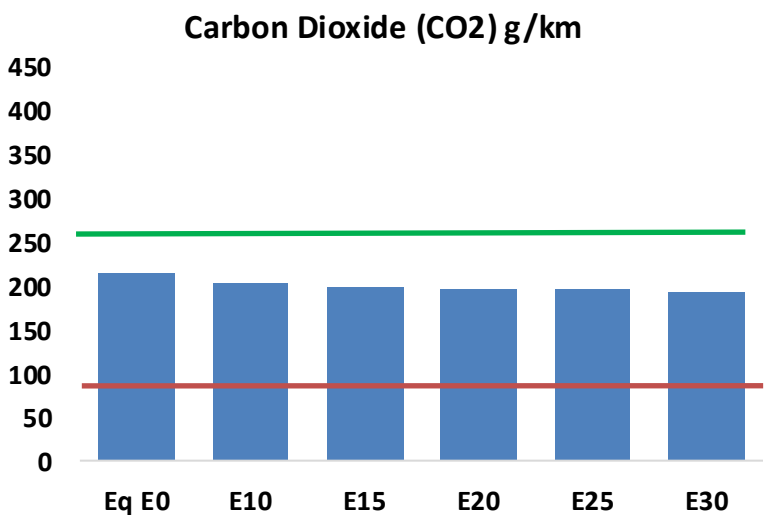
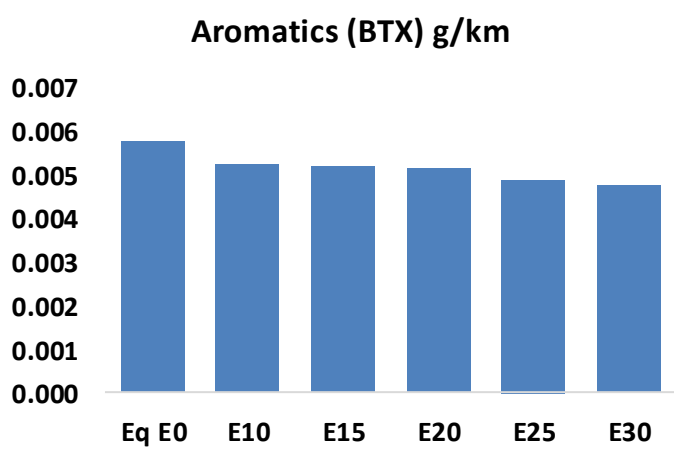
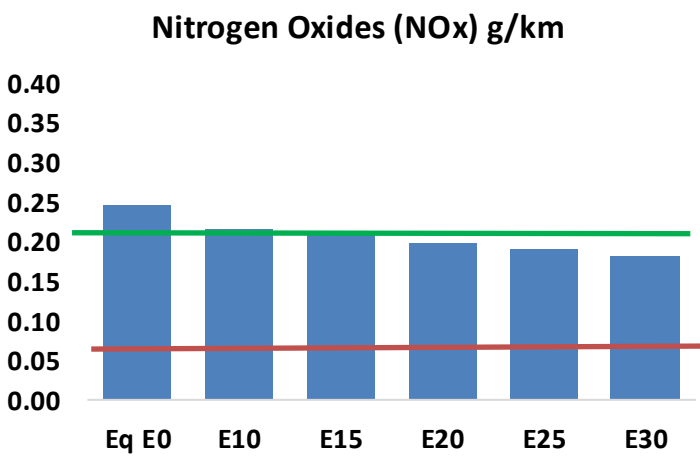
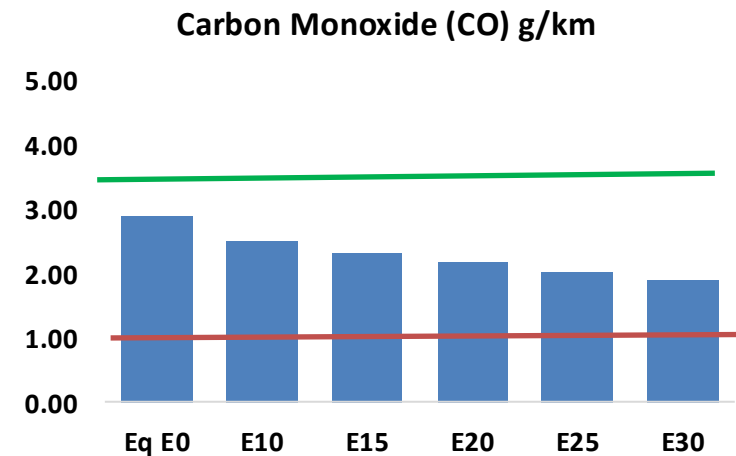
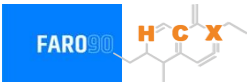


Vehicle Fleet: 36,100,582
Gasoline Vehicle Fleet: 11,736,365

Source: Eurostat, ACEA

Spain – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Spain – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	860,684	799,787	738,890	690,015	641,305	603,992	562,391	-14%	-25%
CO2	62,968,337	61,349,152	59,819,921	58,617,225	58,022,085	57,682,674	56,619,416	-5%	-8%
VOC	127,604	118,925	110,246	103,354	96,520	91,295	84,216	-14%	-24%
NOx	73,232	67,617	63,767	61,807	58,647	56,172	53,256	-13%	-20%
SOx	717	663	609	563	519	471	421	-15%	-28%
NH3	22,043	21,824	21,606	21,709	21,953	22,125	22,267	-2%	0%
N2O	1,010	1,005	1,001	1,006	1,026	1,038	1,050	-1%	2%
CH4	27,368	27,368	27,368	27,778	28,381	28,901	29,393	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	837	810	782	767	754	746	731	-7%	-10%
Acetaldehyde	2,230	2,560	3,756	5,720	7,793	8,905	10,522	68%	249%
Formaldehyde	7,769	7,554	8,805	10,339	10,832	11,983	13,045	13%	39%
Benzene	1,710	1,643	1,557	1,532	1,519	1,453	1,406	-9%	-11%
PM2.5	3,691	3,356	3,020	2,725	2,436	2,126	1,802	-18%	-34%
PM10	4,385	3,986	3,588	3,237	2,894	2,526	2,141	-18%	-34%

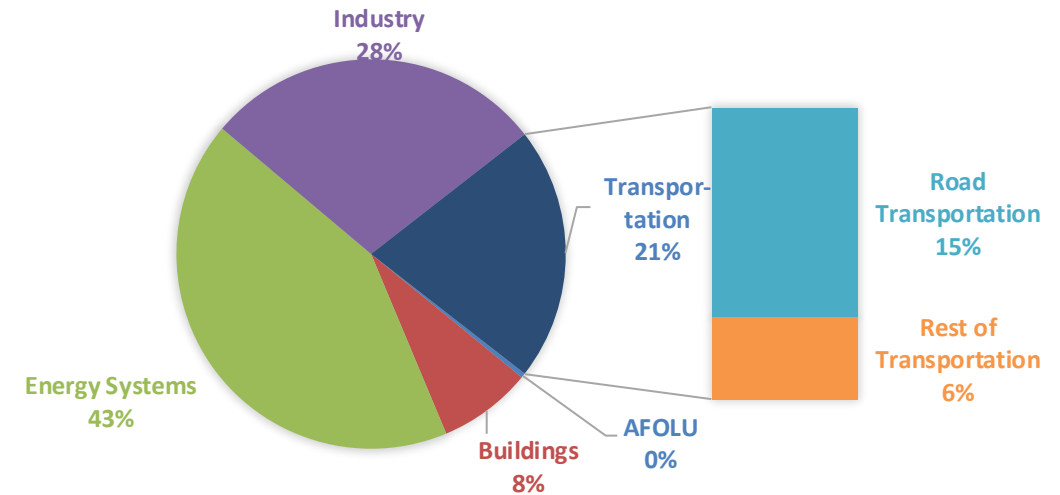
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

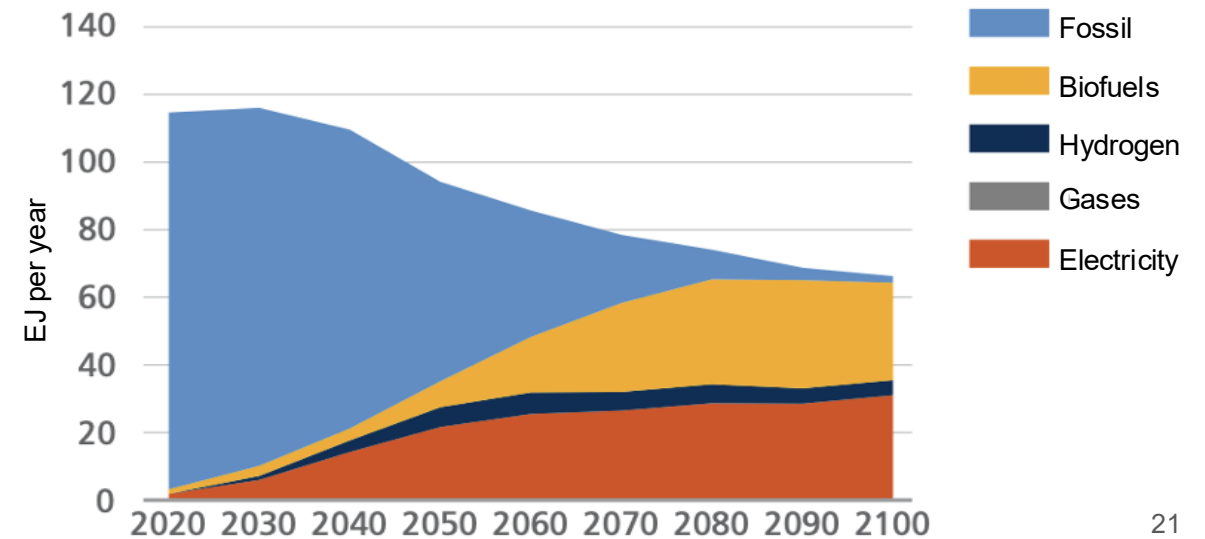
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

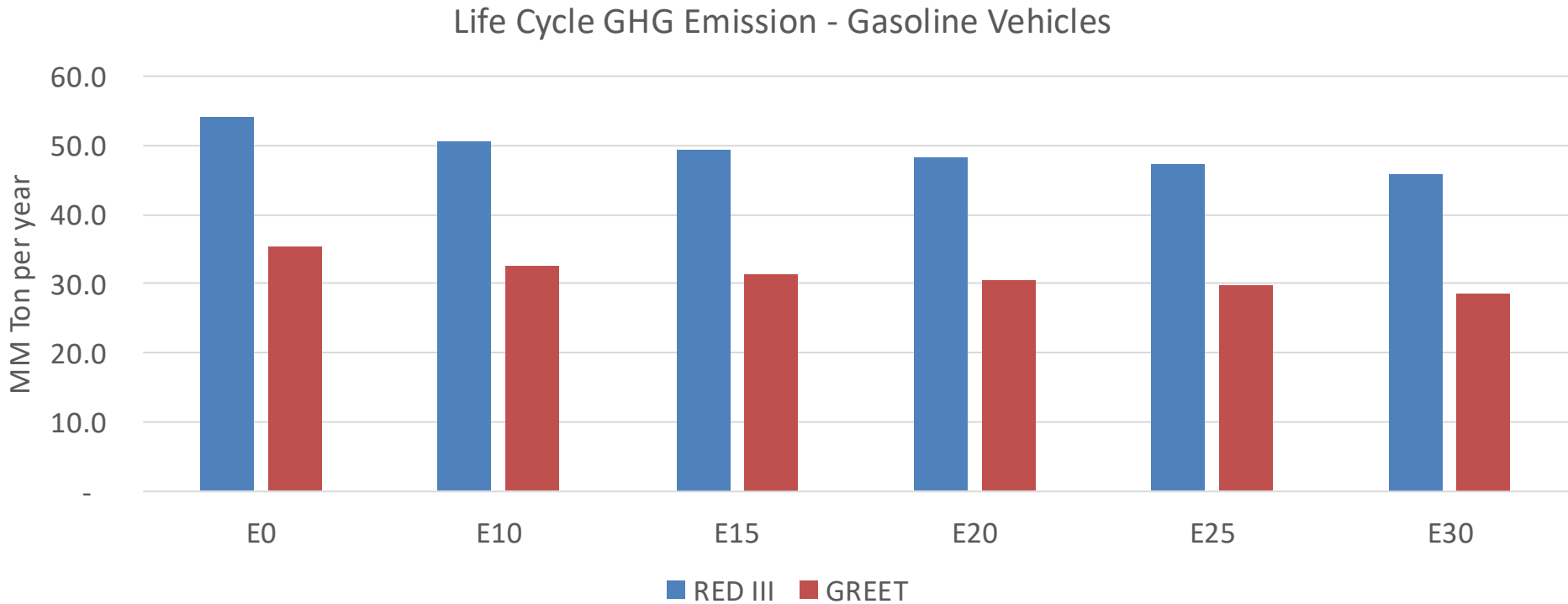
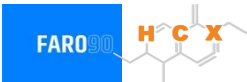
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Spain – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	269.9
Target @ 2035 (MMT/year)	151.9
Delta	118.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.2%	11.2%	13.9%	16.1%	19.3%
RED II/III	0.0%	6.3%	8.6%	10.7%	12.5%	14.9%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.5%	3.3%	4.2%	4.8%	5.8%
RED II/III	0.0%	2.9%	3.9%	4.9%	5.7%	6.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90